

Teaching Exam

Subject – Computer Science

Emerging Trends



Lecture No -2

By-Satyendra Chaudhary Sir

Topics to be Covered

Topic

IOT

IOT → Rest, web sockets

Cloud computing → Architecture.



IoT stands for Internet of Things. Connecting everyday things embedded with electronics, software, and sensors to the internet enables them to collect and exchange data.

Benefits of IoT:

- Efficient resource utilization
- Minimizing Human effort
- Saves time
- Development of AI through IoT
- Improved security

IoT Features:

- Connect various things to the IoT Platform
- Analyze the data collected and use it to build business intelligence
- Integrate various models to improve user experience.

IoT Ecosystem:

Basic architecture:

- **Application layer:** Responsible for delivering application-specific services to the user. Defines an application in which the IoT can be deployed.
- **Network layer:** Responsible for connecting smart things, network devices, and servers. Also used for transmitting and processing sensor data.
- **Perception layer:** Sensors sense and gather information about the environment. Senses physical parameters or identifies other objects in the environment.

Application of IoT: } Sensory

- Energy conservation
- Smart Transport
- Social life and entertainment
- Supply chain and logistics
- Health and fitness
- Smart environment
- Home automation
- Smart agriculture

Supply chain Management

IoT Sensor:

It is a device that captures real-world data and translates it into a piece of information that could be interpreted by other instruments. As the name suggests, sensors react to stimuli – be it changes in temperature, sound, or different properties. Hence a sensor in an IoT system senses the desired physical quantity and converts it into electrical signals transmitted to the central cloud-based server directly or via an on-site microcontroller.

Types of Sensors:

- Temperature sensor
- Humidity sensor
- Proximity sensor
- Level sensor
- Gyroscope.

1. What is the full form of IoT?

- a) Internet of Technology
- b) Incorporate of Things
- c) Internet of Things
- d) Incorporate of Technology

2. What is IoT?

- a) network of physical objects embedded with sensors
- b) network of virtual objects
- c) network of objects in the ring structure
- d) network of sensors

3. Who coined the term "Internet of Things"?

- a) Kevin Aston
- b) John Wright
- c) Edward Jameson
- d) George Garton

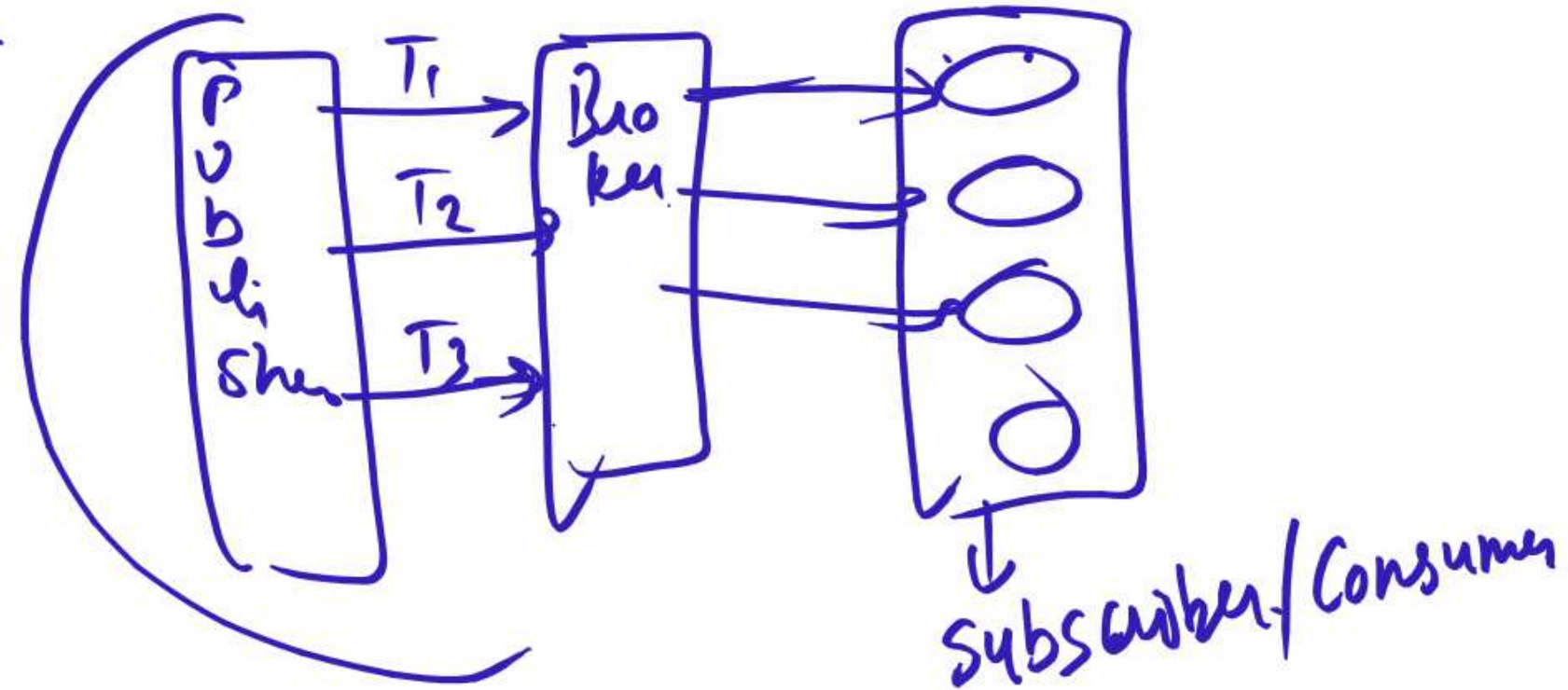
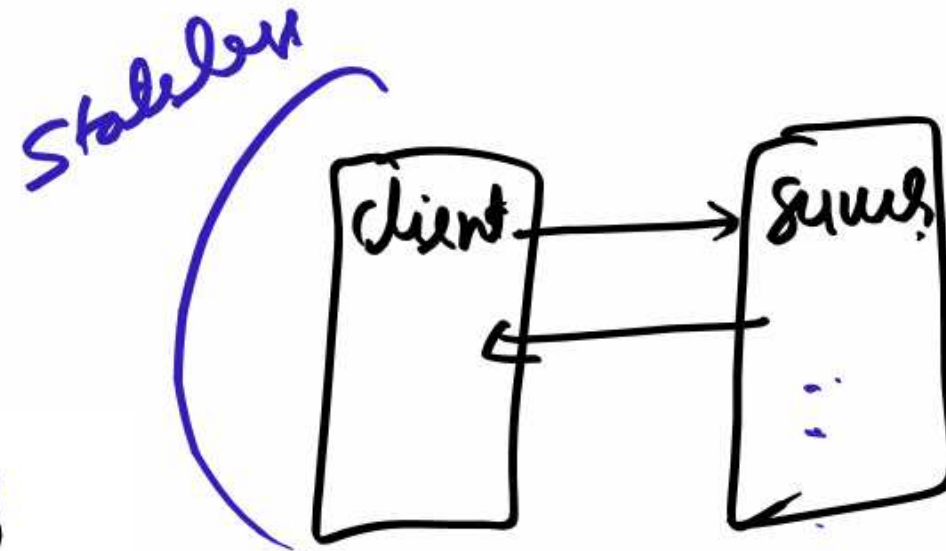
Communication us IoT

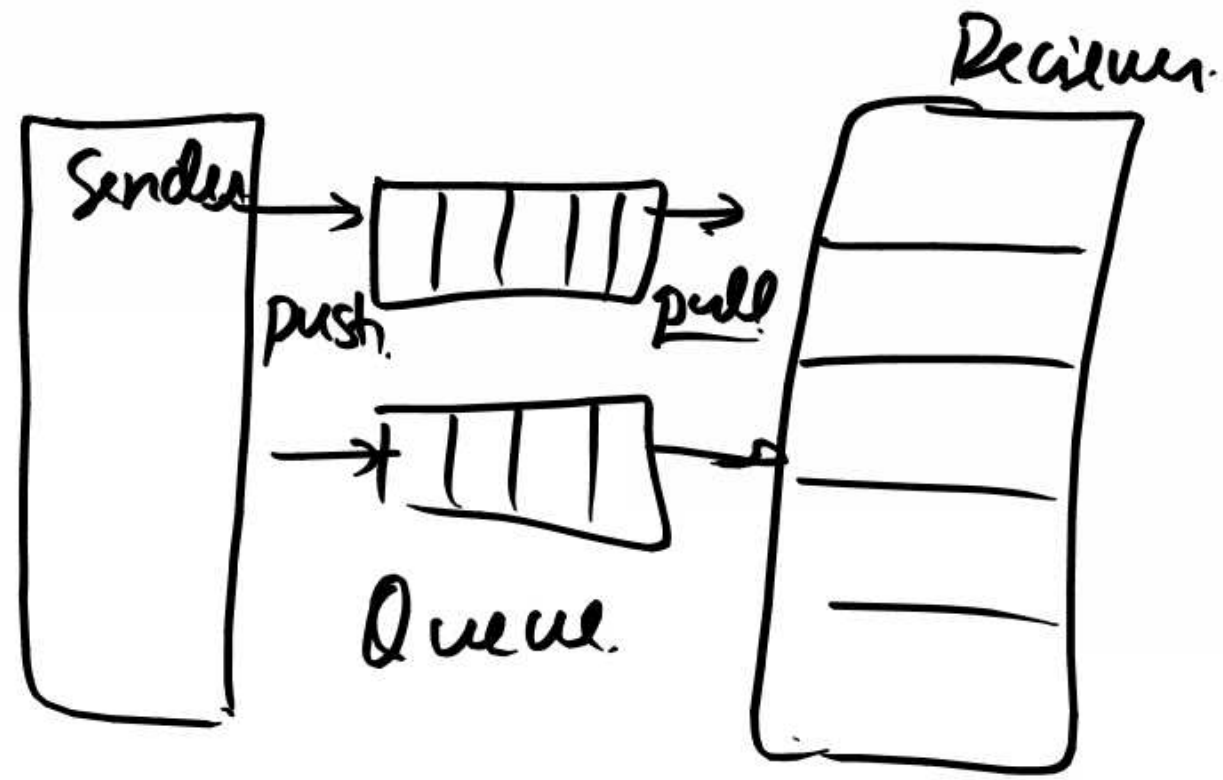
→ Request Response Model.

→ Publisher Subscriber Model.

→ Push Pull.

→ Envelope Pair Model.

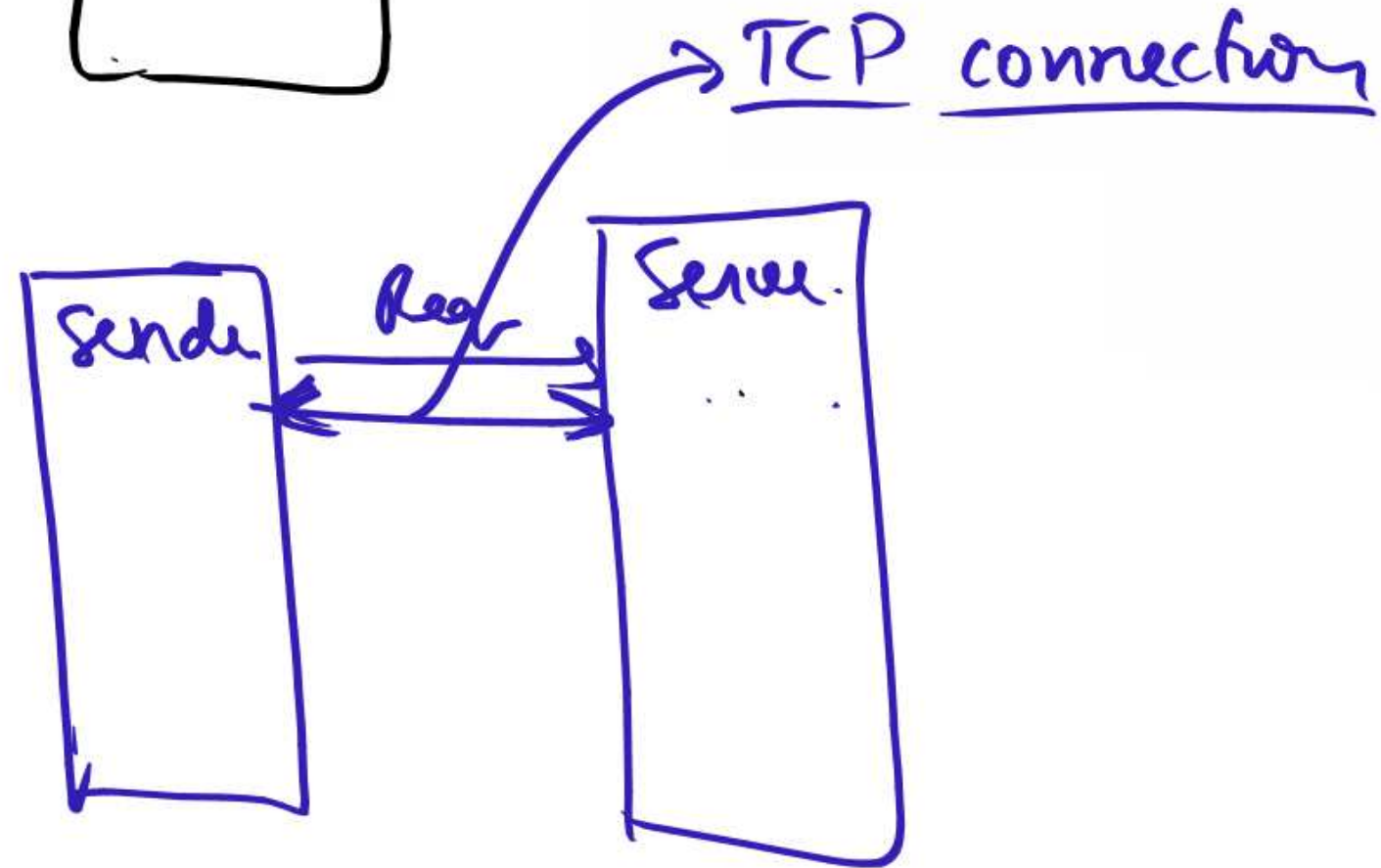




S 7

Exclusive Pair Model

Model



REST,

Stateless

Not full duplex

Web Socket.

Stateful.

Full duplex.

protocol.

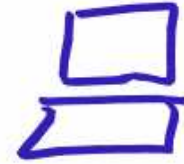
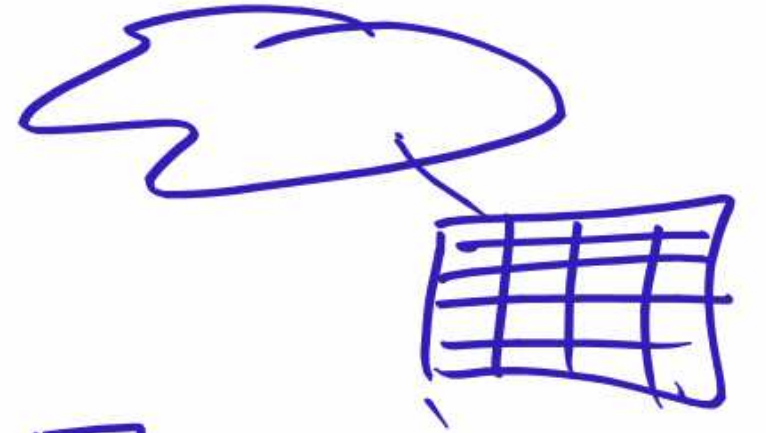
Cloud Computing

→ IaaS → Amazon EC2
↓ → Google compute Engine

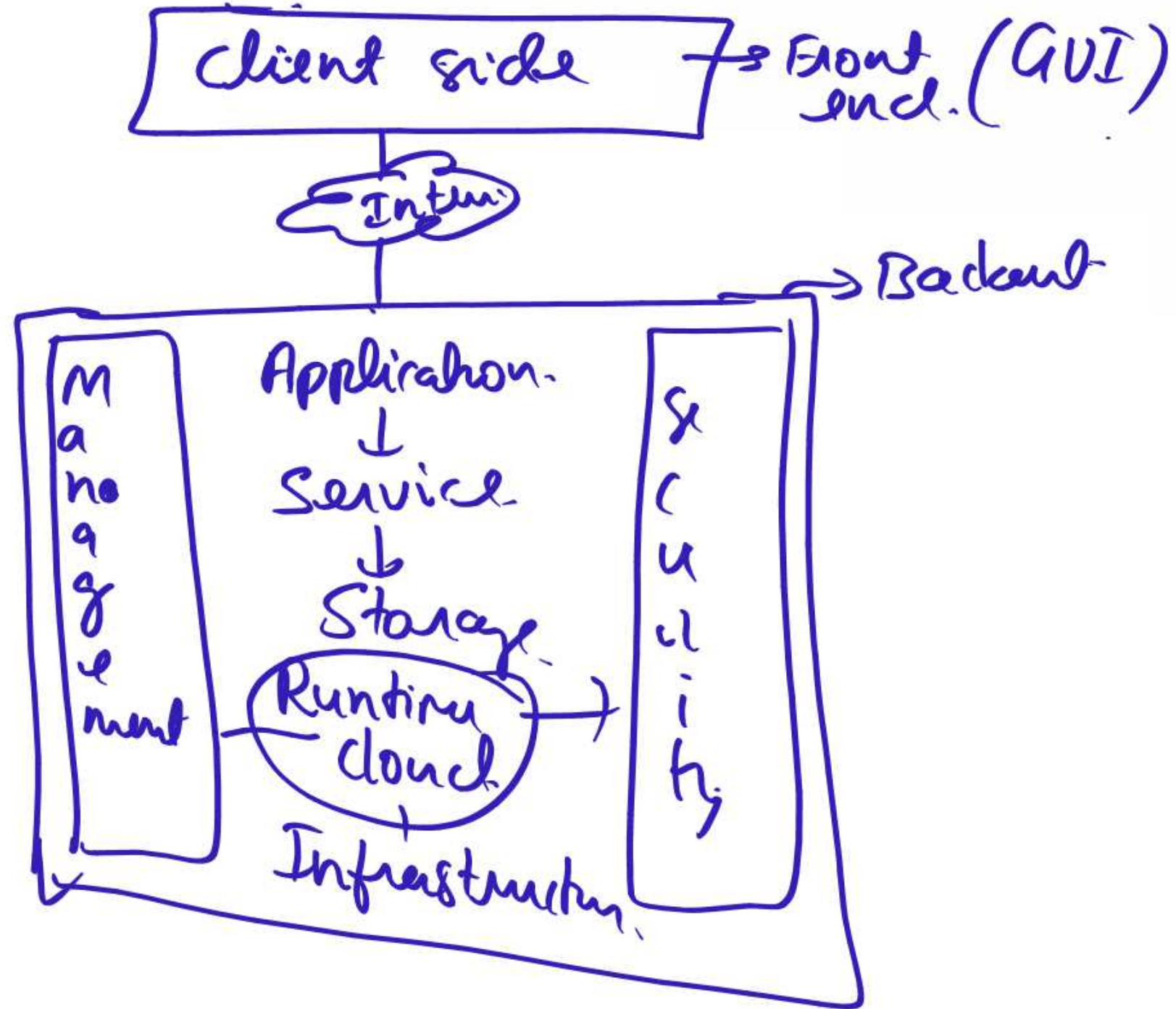
Infrastructure as a service.

Platform as a service (PaaS) → Microsoft Azure
AWS

Software as a Service (SaaS) → Salesforce, Dropbox



Architecture



- Scalability
- ⇒ Upgrade
- ⇒ Cost effective
- ⇒ Security.



2 mins Summary



TEACHING
WALLAH

Topic

IOT

Cloud Computing

THANK YOU